ALICE L WALTON SCHOOL OF MEDICINE

Principles of Chemotherapy

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Adapted from: Cancer Therapy: Foundations and Frameworks

Listen to this Brick

0:00 / 15:23 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Apply the principles of chemotherapy treatment.
2. Explain how the log-cell kill hypothesis guides cytotoxic chemotherapy treatment regimens.
3. Recognize common adverse effects of drugs commonly used in cancer therapy.

CASE CONNECTION

MR is a 56-year-old man recently diagnosed with diffuse large B-cell lymphoma. He begins his first cycle of combination chemotherapy. After the first round, follow-up imaging shows a significant decrease in tumor size, but the tumor is still detectable. MR expresses disappointment, saying, "If the chemo is working, shouldn't it have killed *all* of the cancer cells by now? Why do I need so many more rounds?"

What will you tell MR about how chemotherapy works, specifically in relation to the log-cell kill hypothesis? Consider your answer as you read, and we'll revisit MR at the end of the brick.

GO TO CONCLUSION ↓

What Are the Principles of Chemotherapy? (LO#1)

Cancer pharmacotherapy includes systemic drug therapies targeted to destroy cancer cells without excessive damage to normal cells.

Anticancer Drugs

Anticancer drugs work by a number of mechanisms, including destroying cancer cells with cytotoxins, targeting specific cell receptors and molecular pathways (molecularly targeted therapy, which includes signal transduction inhibitors and certain immunotherapies), and blocking or regulating certain hormones (endocrine a disease of abnormally proliferating cells, the goal of pharmacotherapy is to halt the proliferation of these cells.

Table 1

Therapy	Mechanism of action	Example
Cytotoxic chemotherapy	Uses cellular poisons to prevent cell growth and replication	Methotrexate: used for rheumatoid arthritis and some cancers
Molecularly targeted therapy	Targets specific cellular receptors and molecular pathways	Imatinib: used for chronic myelogenous leukemia
Immunotherapy	Helps the immune system attack the cancer cells, including blocking the protective mechanism of cancer cells, allowing immune cells to destroy the cancer	Pembrolizumab: used for many advanced cancers, including melanoma and cervical cancer
Endocrine therapy	Treats specific hormone-responsive cancers by blocking or modulating certain hormones	Tamoxifen: used for estrogen receptor–positive breast cancer

Timing of Chemotherapy

Chemotherapy can also be classified based on the timing of the therapy:

- **Induction therapy:** First-line, or best, treatment for a particular cancer to kill as many cancer cells as possible
- **Neoadjuvant therapy:** Given before the primary treatment to shrink both the primary cancer (making it easier for the surgeon to resect) and distant micro-metastase
- **Adjuvant therapy:** Given after the primary treatment to improve outcomes
- **Salvage therapy:** Used when all preferred therapies have been tried and have failed to produce cancer remission

What is the purpose of neoadjuvant therapy in cancer treatment?

Measuring the Effect of Chemotherapy

Another important term used in discussing chemotherapy drugs is **progression-free survival**, the period after the treatment in which a patient's disease does not get worse. This is not the same as **overall survival**, which refers to the length of time after treatment that the patient lives.

Drug Combinations vs Drug Monotherapy

Cancer, just like bact to drugs in a variety of ways. One particularly common mode of resistance is DNA mutations. For example, a mutation in a particular receptor target or enzyme may prevent a drug from binding. Drug combinations can be used to combat this resistance. This way, even if a cancer cell develops resistance to one drug, another drug may be able to kill it.

One reason cancers may develop resistance is that cancer cells are not usually a uniform population but have many genetic variants, some of which resist the drug and proliferate. However, even if originally sensitive to the drug, some cancers can become resistant over time. Explanations for why this might happen include:

- Cancer cells may resist uptake of the drug because the protein that transports the drug across the cell membrane stops working.
- Cancer cells may pump the drug out of the cell as fast as it enters, using the multiple drug–resistance protein (MDR), a membrane transporter that can shuttle drugs out of the cell. This MDR upregulation can be a direct mutation or be induced by cytokines like interleukin 6 (IL-6) produced by the tumor cells.
- Some of the cancer cells that are not killed by the chemotherapy mutate and become resistant to the drug.
- The cancer cells upregulate DNA-repair enzymes, correcting the breaks induced by the drug.
- Cancer cells undergo rapid chromosomal rearrangement, leading to amplification or deletion of genes. This may diminish the drug's effect.
- Cancer cell DNA can undergo increased or decreased methylation, which confers decreased drug sensitivity.
- Increased intracellular expression of **glutathione** allows it to react with chemotherapy dr

How might cancers resist chemotherapy drugs?

What is the log–cell kill hypothesis? (LO#2)

Chemotherapy follows the **log-cell kill hypothesis**, which states that a given dose of chemotherapy kills a constant *proportion*—rather than a constant number—of cancer cells. This reflects first-order kinetics: the magnitude of tumor cell reduction is logarithmic. For example, if a tumor has 10^{12} cells, a 3-log kill would reduce the burden to 10^9 cells; similarly, if a smaller tumor has 1,000 cells and 95% are eliminated with each dose, about 50 cells would remain after the first dose and two after the second (Figure 1). Because no matter how small the tumor is, a single treatment rarely eradicates all cells, **multiple cycles of chemotherapy are required** (Figure 2) to shrink the tumor to a clinically insignificant size.

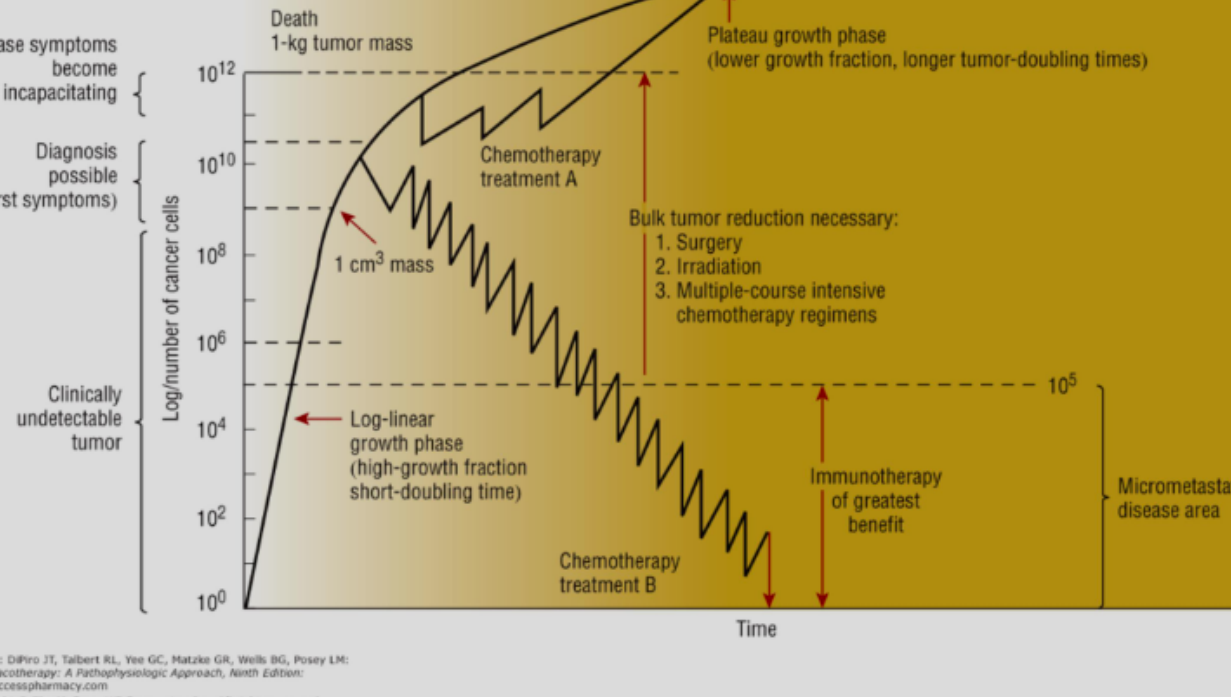


Figure 1

This principle is most applicable to rapidly dividing malignancies such as acute leukemias and aggressive lymphomas. In solid tumors, growth rates slow as tumors enlarge due to limitations in blood supply, meaning treatment responses can be more variable. Importantly, chemotherapy must balance killing malignant cells while sparing enough of the patient's healthy, rapidly dividing cells. Thus, treatment is given in cycles: an intensive drug dose reduces tumor burden, followed by a recovery period that allows normal tissues to regenerate, before another cycle resumes the process. Ultimately, **tumor cell burden and growth kinetics** are major determinants of chemotherapy scheduling and success.

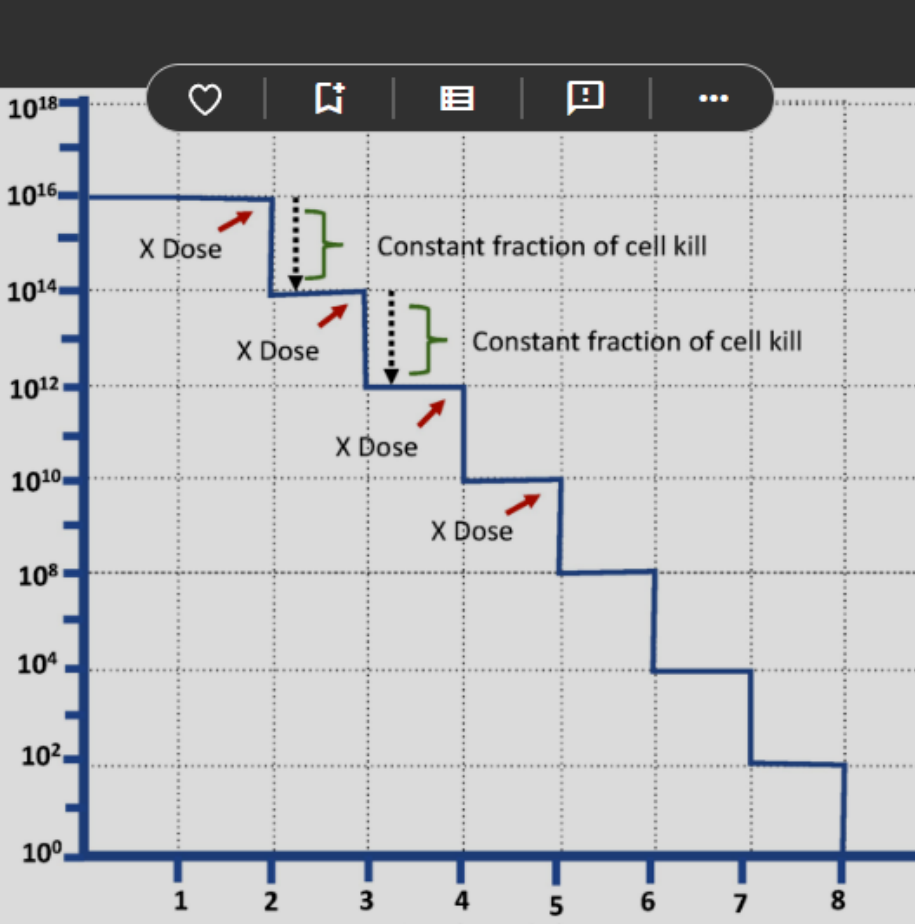
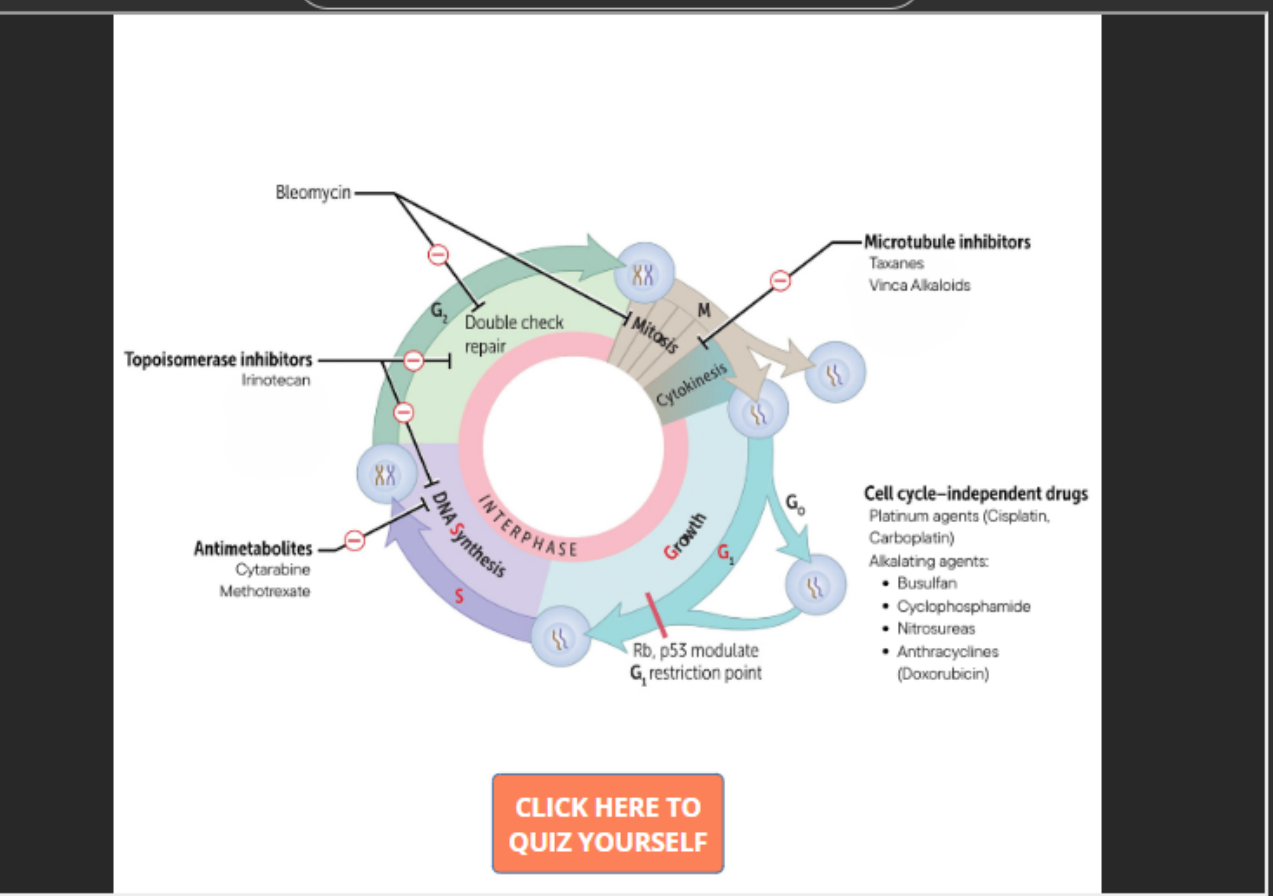


Figure 2

How do anticancer drugs work?

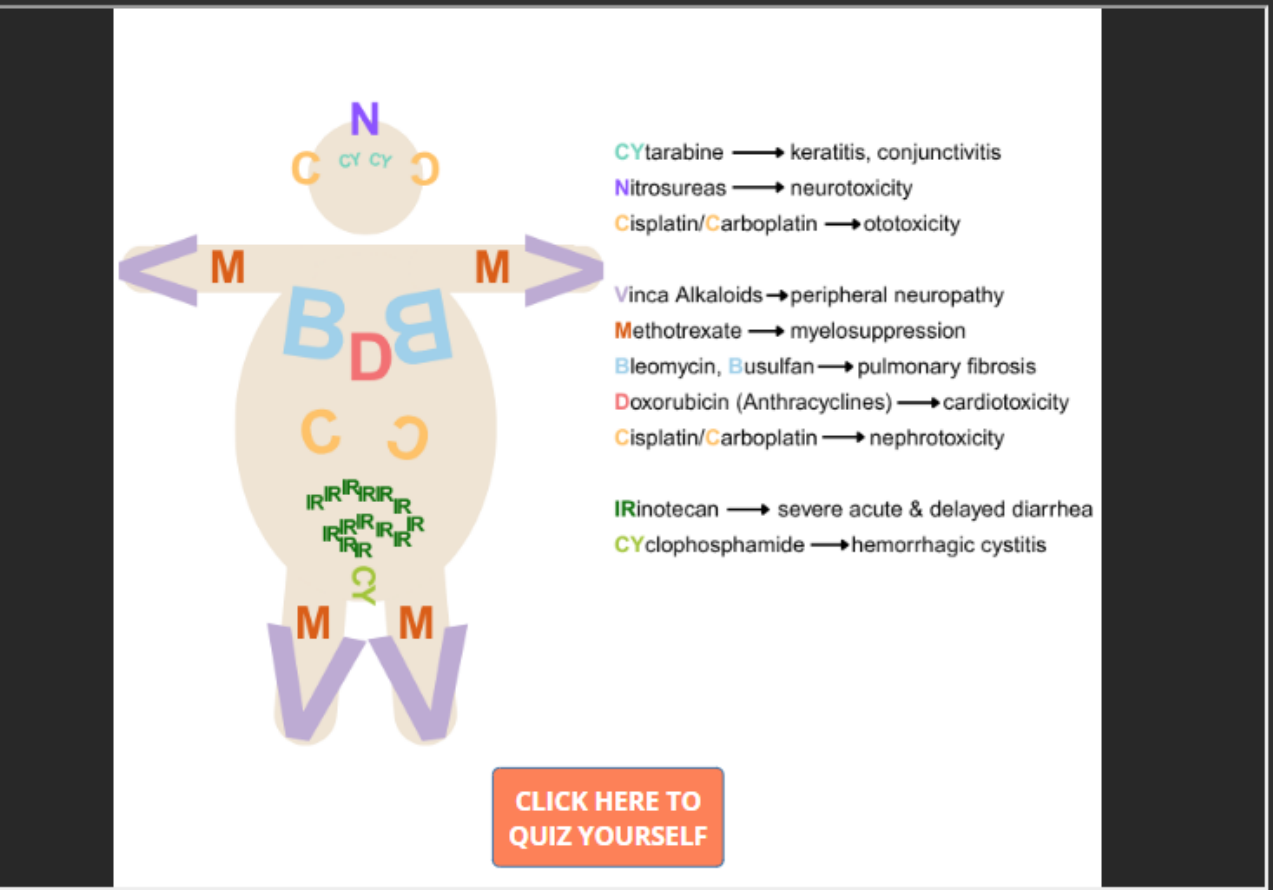
Above we mentioned the that anticancer drugs work by a number of mechanisms. In the 'Cancer Medications Overview' Brick from the Foundations of Pathology course, we focused on traditional cytotoxic chemotherapy. These drugs inhibit different parts of the cell cycle. Quiz yourself on the image below to see how well you remember these medication class from previously covered material. These drugs will be a major focus of the pharmacological treatments we discuss in the coming coursework, and we will have individual bricks focused on specific drug classes.



We have also previously mentioned targeted therapies (e.g., monoclonal antibodies) and immunotherapies (e.g., CAR-T). Since we will be discussing these therapies in the context of the cancers they treat we will not detail mechanisms of them here in this brick.

What Are the Adverse Effects of Chemotherapeutic Drugs? (LO#3)

Do you remember ChemoMan? As we previously discussed chemotherapy drugs target any rapidly divi also lead to the death of normal cells that t ll have varying affinities for rapidly dividing cells throughout the body. So, drugs that affect the gastric mucosa will likely lead to nausea and vomiting, drugs that affect bone marrow cells will result in myelosuppression, etc. In the coming weeks we will be covering the specific classes of chemotherapy drugs, these adverse effects will show up again and there will be more information on other adverse effects. Again, this ChemoMan image is very important to be familiar with as these effects are very common and highly testable!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CYtarabine → keratitis, conjunctivitis
Nitrosureas → neurotoxicity
Cisplatin/Carboplatin → ototoxicity
Vinca Alkaloids → peripheral neuropathy
Methotrexate → myelosuppression
Bleomycin, Busulfan → pulmonary fibrosis
Doxorubicin (Anthracyclines) → cardiotoxicity
Cisplatin/Carboplatin → nephrotoxicity
IRinotecan → severe acute & delayed diarrhea
CYclophosphamide → hemorrhagic cystitis
CLICK HERE TO QUIZ YOURSELF

In general, the more specific agents like monoclonal antibodies and small molecule inhibitors r targeted way, which reduces the number and intensity of adverse effects.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to MR, what will you tell him about why multiple rounds of chemotherapy are needed?

You explain that chemotherapy doesn't kill all cancer cells at once, even when it's working well. Instead, each dose kills a fixed fraction of the remaining cancer cells — a concept called the log-cell kill hypothesis. For example, if a drug kills 90% of cancer cells with each cycle, a tumor with one billion cells would be reduced to 100 million after the first dose, then to 10 million after the second, and so on. This means several rounds are necessary to drive the cancer cell count down to a level where the immune system or other treatments can eliminate the remaining cells, or they are too few to cause disease. By completing all planned cycles, MR maximizes the chance of achieving long-term remission.

Summary

- Cancer pharmacotherapy aims to destroy cancer cells while minimizing damage to normal cells.
- Chemotherapy mechanisms include cytotoxic drugs, molecularly targeted therapy, immunotherapy, and endocrine therapy.
- Drug resistance in cancer can occur through genetic mutations, decreased drug uptake, drug ejection, and DNA repair upregulation.
- The log-cell kill hypothesis suggests chemotherapy kills a fixed fraction of cancer cells per c tive treatment.
- Chemotherapy adverse effects arise from targeting rapidly dividing cells, affecting normal cells like those in the gastric mucosa and bone marrow.
- Combination chemotherapy is preferred over monotherapy due to the potential for drug resistance in cancer cells.

Review Questions

1. A patient with non-Hodgkin lymphoma undergoes chemotherapy. After two cycles, imaging shows a dramatic reduction in tumor size, yet some cancer is still visible. He asks why treatment must continue despite the positive response. Which concept best explains the need for multiple cycles?

A. ○ Adjuvant therapy
B. ○ Drug resistance
C. ○ Induction therapy
D. ● Log-cell kill hypothesis
E. ○ Tumor heterogeneity

Hide Explanation

The correct answer is the log-cell kill hypothesis (D). This explains that each cycle of chemotherapy kills a consistent fraction of remaining cells, not all of them at once. Even if imaging shows shrinkage, residual disease requires further treatment to reduce the cell burde do not directly explain the fraction-based killing that necessitates multiple cycles.

2. Which therapy shrinks tumors pre-surgery?

A. ○ Induction therapy
B. ● Neoadjuvant therapy
C. ○ Adjuvant therapy
D. ○ Salvage therapy
E. ○ Targeted therapy

Hide Explanation

The correct answer is neoadjuvant therapy (B). Neoadjuvant therapy aims to shrink the primary cancer and micro-metastases before the main treatment such as surgery. Induction therapy (A) is the first-line treatment for killing cancer cells, not for pre-surgery shrinkage. Adjuvant therapy (C) is given post-surgery to improve outcomes. Salvage therapy (D) is used after preferred therapies have failed. Targeted therapy (E) involves drugs that target specific pathways, not specifically pre-surgical shrinkage.

3. Which chemotherapy agent is associated with keratitis?

A. ○ Bleomycin
B. ● Cytarabine
C. ○ Doxorubicin
D. ○ Methotrexate
E. ○ Vinca Alkaloids

Hide Explanation

The correct answer is cytarabine (B). Keratitis/ conjunctivitis are commonly associated with cytarabine. Bleomycin (A) is most commonly associated with pulmonary fibrosis. Doxorubicin (C) is associated with cardiotoxicity. Methotrexate (D) is associated with myelosuppression and Vinca Alkaloids (E) are associated with peripheral neuropathy.

How would you rate this Brick?

Review Learning Objectives

Check off th the concept

The correct answer is cytarabine (B). Keratitis/ conjunctivitis are commonly associated with cytarabine. Bleomycin (A) is most commonly associated with pulmonary fibrosis. Doxorubicin (C) is associated with cardiotoxicity. Methotrexate (D) is associated with myelosuppression and Vinca Alkaloids (E) are associated with peripheral neuropathy.

How would you rate this Brick?

Review Learning Objectives ✓

Check off the objectives when you feel comfortable with the concept

- 01 ☑ ~~Apply the principles of chemotherapy treatment.~~
- 02 ☑ ~~Explain how the log-cell kill hypothesis guides cytotoxic chemotherapy treatment regimens.~~
- 03 ☑ ~~Recognize common adverse effects of drugs commonly used in cancer therapy.~~

Objective Confidence: 3/3